

Exam Date & Time: 04-Jan-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru
BTech III Semester - Make up Examination, Jan-2025
Analog Electronic Circuits [ECM-2121]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Design the source follower depicted in Fig. 1A for a voltage gain of 0.8 and a power budget of 2mW. Assume the output dc level is equal to $V_{DD}/2$ and the input impedance exceeds 10 k Ω .

A)

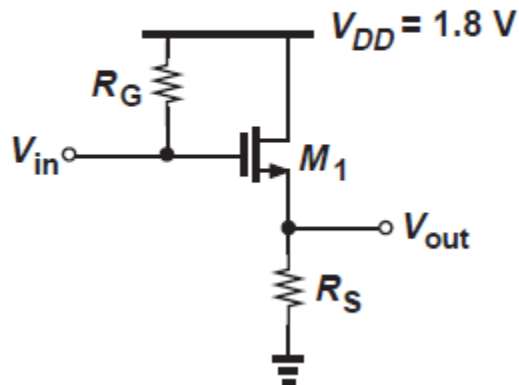


Fig. 1A

(4)

B)

A MOSFET is biased at a drain current of 0.5 mA. If $\mu_n C_{ox} = 100 \mu\text{A}/\text{V}^2$, $W/L = 20$, and $\lambda = 0.1 \text{V}^{-1}$, calculate its small-signal parameters.

(3)

C)

The PMOS cascode of Fig. 1C must provide a bias current of 0.5mA with an output impedance of 40 k Ω . If $\mu_p C_{ox} = 50 \mu\text{A}/\text{V}^2$ and $\lambda = 0.2 \text{V}^{-1}$, determine the required value of $(W/L)_1 = (W/L)_2$.

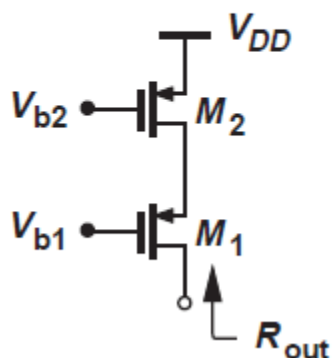


Fig. 1C

(3)

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(3)

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- The diagram shows a CMOS inverter circuit. The input node is labeled 'X' and is connected to a voltage divider consisting of a $4\text{ k}\Omega$ resistor (R_1) in series with a $10\text{ k}\Omega$ resistor (R_2), both connected to ground. The output node is labeled 'Y' and is connected to a drain resistor R_D in series with a source resistor R_S (labeled $1\text{ k}\Omega$), both connected to ground. The supply voltage is $V_{DD} = 1.8\text{ V}$. The NMOS transistor is labeled M_1 and has its gate connected to node 'X' and its source connected to node 'Y' through R_S . The PMOS transistor is represented by a downward arrow symbol with its gate connected to node 'X' and its source connected to V_{DD} . The drain current is labeled I_D .

(3)

- (3)

For the circuit shown in Fig. 3B find the pole frequencies. Assume $\lambda = 0$, $g_m = 150\Omega^{-1}$, $V_{DD}=1.8V$, $R_S=200\Omega$, $R_D=2K\Omega$, and $C_F=80\text{ fF}$, calculate C_{in} & C_{out} .

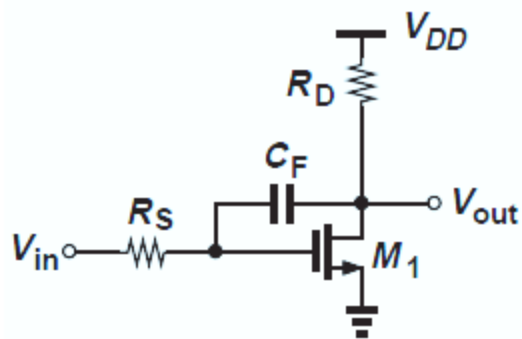


Fig. 3B

- C) In the amplifier of Fig. 3C, $R_D = 1\text{ k}\Omega$ and $C_L = 1\text{ pF}$. Neglecting channel-length modulation and other capacitances, determine the frequency at which the gain falls by 20%? ($\approx 2\text{ dB}$).

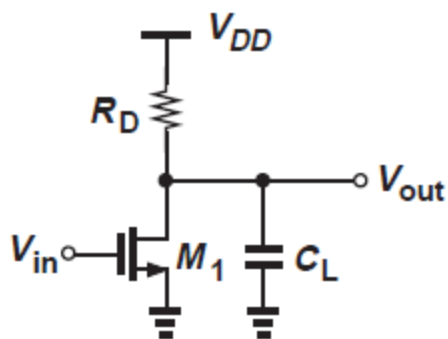


Fig. 3C

- 4) Identify all the capacitances and reduce the circuit shown in Fig. 4A to compute the poles. Assume $\lambda = 0$.

A)

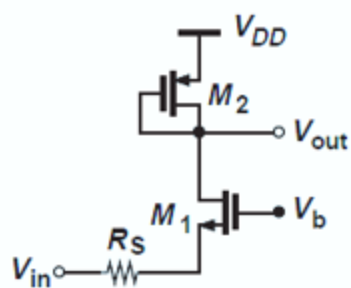
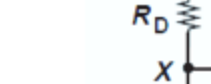


Fig. 4A

B)

A) 

(3)

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